**TRANSFORMERS QUESTIONS AND ANSWERS**
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## 1) What is a Transformer ?

**Ans :** A transformer is a static piece of apparatus which transforms electrical energy from one Circuit to another circuit without changing the frequency either stepping up or stepping down the voltage .

## 2 .On what theory the principal of operation of a transformer is based ?

**Ans :** The principle of operation of a transformer is based upon the theory of mutual induction Between two electrical linked by a common magnetic flux .

## 3 . What do you understand by the rating of a transformer ?

Ans: The rating of a transformer is the maximum power which can be drawn from it without the temperature rise in the winding exceeding the safe limits for the particular class of insulation employed .

## 4.How is the rated capacity of a transformer expressed and why ?

Ans: The rated capacity of a transformer is expressed in KVA not in KW. Generally the rating of a transformer is determined by its temperature rise . The temperature rise is caused by the losses in the machine . Copper loss depends on the value of the load current and iron loss on voltage . Therefore ,total loss of a transformer depends on volt-ampere (VA) and is independent of load power Factor. A certain amount of current will produce the same $I^2$ R loss at any value of power factor . This loss limits the output of the machine .The output in kilowatts is proportional to the power factor . For a given KW load if the power factor decreases, the load current increases proportionately causing more losses and temperature rise in the machine . For the above reason, transformers are generally rated in KVA, not in KW .

## 5. What is the power factor of a transformer ?

Ans : The power factor of a transformer is very low and lagging on no load . But the power factor on load is nearly equal to the power factor of the load which it is carrying .

## 6. What is the normal phase difference between the voltage and the on load current in a transformer ?

Ans :The no load current in a transformer normally lags behind the voltage by about $7^0$.

## 7.What are the essential parts of a transformer ?

Ans: The essential parts are as follows :- a)Magnetic circuit consisting of laminated iron core and clamping structures. b)Primary winding . c)Secondary winding . d)Tank filled with insulating oil . e)H.T terminals with bushing . f)L.T terminals with bushing . g)Conservator tank. h)Breather. i)Vent pipe . j)WTI ( Winding Temperature Indicator ) & OTI ( Oil Temperature Indicator). k)Radiator.

## 8.What is the name of the winding to which the supply is given ?

Ans : Primary winding .

## 9 . What is the name of the winding from which the supply is taken for load connecting ?

Ans: Secondary winding .

## 10. What material is used for the cores of a transformer and why ?

Ans: Lamination of specially alloyed silicon steel (silicon content 4 to 5%) are used due to its high electrical resistance ,high permeability non-ageing characteristics and minimum iron loss .

## 11. What is the use of iron core in a transformer ?

Ans :The iron core is used in a transformer to provide a continuous easy magnetic path of low reluctance .

## 12.How is magnetic leakage reduced ?

Ans: The magnetic leakage is reduced to a minimum by sectionalizing & interleaving the primary and secondary windings .

## 13. Why should the joints in iron core be staggered ?

Ans: The joints in iron core should be staggered to avoid clear air gap in the magnetic circuit as the air gap reduces the magnetic flux due to its high reluctance .

## 14.Why are iron cores made laminated in the transformer ?

Ans: Iron cores are made laminated to reduce the eddy current loss .

15. What determines the thickness of the lamina or stamping ?

Ans: Frequency.

16.Why is the lamination insulated from each other ?

Ans: The laminations are insulated from each other by insulating varnish or thin paper to break the path of eddy currents thus to reduce the eddy current loss .

17. What is stacking factor ? what is its approximate value ?

Ans: Stacking factor is the ratio of iron content in the laminated varnished core by volume. Its value is about 90% i.e. 10% volume is occupied by the varnish and air .

18. What is called grain oriented laminations ?

Ans: The grain oriented laminations are such cold rolled laminations specially annealed to orient the iron crystals i.e the grains in a uniform direction in the direction of rolling which produces loss . These laminations are punched and assembled in such a way that the grains of different lamina line in the same direction as the magnetic field and not across it .

19. What is the permissible maximum flux density in transformer core ?

Ans: 1.6 to 1.8 weber per meter .

20. Why is special care taken during tightening of iron cores ?

Ans : To make it free from vibration and humming sound.

21. What is the phase relationship between the primary and secondary voltages of a transformer?

Ans:$180^o$ out of phase .

22. What is called turn ratio of a transformer ?

Ans: The ratio of number of turns in the primary to the number of turns in the secondary winding is called the turn ratio or the ratio of transformation of the transformer which is indicated by a constant 'K' .Than $(N_1/N_2) = E_1/E_2 = K$ .

## 23. What is a voltage ratio of a transformer?

Ans : Voltage ratio is the ratio of the voltage between the line terminals of one winding to that between the terminals of another winding at no load .

## 24. What is the relation between voltage ratio and turn ratio of (a) D/D ,(b) D/Y,(c) Y/Y &(d) Y/D transformer ?

Ans: For Δ/Δ transformer : $\frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2}$

For Δ/Y transformer: $\frac{V_1}{V_2} = \frac{E_1}{\sqrt{3}\,E_2} = \frac{N_1}{\sqrt{3}N_2}$

For Y/Y transformer: $\frac{V_1}{V_2} = \frac{\sqrt{3}E_1}{\sqrt{3}E_2} = \frac{N_1}{N_2}$

For Y/Δ transformer: $\frac{V_1}{V_2} = \frac{\sqrt{3}E_1}{E_2} = \frac{\sqrt{3}N_1}{N_2}$

## 25. What do you mean by true ratio of transformation and commercial ratio transformations ?

Ans: The true ratio of transformation is the ratio of phase voltages of the primary and secondary winding whereas the commercial ratio of transformation is the ratio of the primary and secondary line voltages .

## 26. What is the relation between primary ampere-turns to the secondary ampere-turns ?

Ans : The primary and secondary ampere-turns are equal and opposite i.e. $I_1 N_1 = I_2 N_2$ Then $I_1/I_2 = N_2/N_2$ i.e the primary and secondary currents are inverse as the respective turns .

## 27. How is the secondary voltage calculated by the turns of the winding ?

Ans: The voltage per turn is practically constant in particular transformer so the secondary voltage is calculated by multiplying the input primary voltage per turn to the number of secondary turns .

## 28.What current flows in the transformer primary when its secondary is open ?

Ans: When the secondary is open there is no current in the secondary of the transformer. The primary takes a small current ($I_0$)from the source called as no load current which has two components, a **magnetising component** ($I_0 \sin \phi_0$ )producing

the magnetic flux and a working component ($I_0 \cos \emptyset_0$ ) supplying real power for iron losses .

## 29. What is the formula for calculating no load current ?

Ans: No load current, $= \sqrt{(I_\omega)^2 + (I_\mu)^2}$

$= \sqrt{(I_0 \cos \emptyset_0)^2 + (I_0 \sin \emptyset_0)^2}$

## 30. Why is the frequency not changed during transformation of electrical energy in a transformer ?

Ans: As the same flux having a definite frequency is responsible for the production of e.m.fs on both the primary and secondary windings so there is no question of changing the frequency .

## 31. How is the induced e.m.f related with the flux linkage ?

Ans : The induced e.m.f is directly proportional to the flux linkage .

## 32. What do you mean by the counter e.m.f in the primary winding of a transformer ?

Ans :when an alternating voltage to the primary winding it causes a flow of alternating current which creates alternating flux. This alternating flux is linked with both the primary and secondary winding and hence induces self-induces e.m.f on the primary winding and a mutually-induced e.m.f on the secondary winding . Since the primary self-induced e.m.f is in the opposite direction to the applied voltage , it is known as counter e.m.f of the transformer .

## 33.What is the e.m.f equation of a transformer ?

Ans: $E_{(r.m.s)} = 4.44\ \emptyset_m.f.T.$ volts Where $\emptyset_m$=maximum value of the flux linking with both the winding . F=frequency of the supply. T=number of turns.

## 34. What are the types of transformers according to transformation?

Ans: (1) Step up transformer & (2) Step down transformer.

## 35. What are the types of the transformer according to the arrangement of iron cores ?

Ans : there are three types :- (a) Core type (b) Shell type (c) Distributed core or Berry type.

## 36. What is the difference between a core type and a shell type transformer?

Ans :In core type transformer the primary and secondary winding are placed on each leg having one magnetic circuit while in shell type transformer both the winding are placed on a common central core having two magnetic circuits. In core type transformer the iron loss will be less and copper loss will be more as compared with a shell-type transformer .

## 37. What magnetic circuit is formed in Berry type construction and why ?

Ans : Distributed magnetic circuits are formed in Berry type construction because of its distributed cores.

## 38. What is called limb?

Ans: the vertical portion of the iron core where the windings are placed is called limb.

## 39. Why are the primary and secondary windings placed on each limb in core type transformer ?

Ans: To reduce the leakage flux.

## 40. Why is L.T winding placed near the cores?

Ans: To reduce the total dielectric strength of the insulating materials provided on the winding.

## 41. What is meant by leakage reactance of a transformer ?

Ans : Practically the flux of primary winding does not link fully the secondary winding . A portion of primary flux without linking the secondary winding passes through the air path. This flux opposes the main circuit current and behaves in the same way as in inductance . The effect is same as that of a reactance acting in series to the primary circuit .

## 42. What are the types of transformers according to winding and supply?

Ans : (a) The single phase transformers ,(b) The three phase transformers and (c)The polyphase transformers.

## 43. What are the types of winding according to the construction?

Ans: Sandwich-type and Cylindrical type.

## 44. How is the winding named according to the voltage ?

Ans:High voltage winding and Low voltage winding.

## 45. What is the difference between cylindrical type and sandwich type winding ?

Ans: In cylindrical type winding the length of the coil is equal to the length of the core limb and the primary and secondary windings are placed one over the other placing low voltage winding nearer the core ,while in sandwitch type winding the H.T and L.T winding are placed on limb lengthwise one above the other link the sandwitch.

## 46. Name the different types of coils used in the transformer winding and state there used.

Ans: There are four types of coils e.g.
(a)Spiral type coil - This consists of layers wound in a continuous length from top to bottom of the coils ,the conductors of which consist of a number of square or rectangular strips in parallel. These coils are only used for low voltage winding carrying a heavy current of more than 100 amps .
(b)Cross over type coil - this type of consists of a number of layers having a number of turns per layer, the conductor being round wire or a strip insulated with a paper covering .Cross over coils are wound on formers . These coils are used for high voltage winding of distribution transformer and current up to about 20 amps .
(c) Helical type coil - This type of coil is wound in the form of a helix and consists of a number of rectangular strip wound in parallel radially so that each separate turn occupies the total radial depth of the winding . These coils are used for lower voltage winding of the larger transformer at voltages 11KV to 33KV . Multilayer helix can be used for higher voltage winding up to 500KV.
(d)Continuous disc-type coil – This coil consists of a number of discs wound continuously from a single wire or a number of strips in parallel . In each disc, a number of turns are wound radially over one another and the conductor passes from

disc to disc uninterruptedly . These coils are used for both higher voltage winding from 33KV to 500KV and lower voltage winding from 11KV to 66KV above the rating of 1 MVA.

## 47. What insulation is largely used in the windings of a high voltage transformer ?

Ans: Varnish coated paper insulation is largely used for the winding of a high voltage transformer .

## 48. What do you mean by uniform insulation ?

Ans: Where the insulation to earth at all points of a winding can withstand the separate source test voltage appropriate to the line end then it is called as uniform insulation . This is generally used up to maximum system voltage of 72.5 KV (r.m.s).

## 49. What is meant by graded insulation?

Ans: In high voltage transformers or banks of transformers where the high voltage winding is star connected and permanently and solidly grounded, the insulation to ground is reduced in steps or graded from the high voltage terminal to the grounded terminal .It is called graded insulation and generally used for highest system voltage of 72.5KV (r.m.s) and above.

## 50. What are the types of transformers according to its location ?

Ans:(1) Indoor type transformer & (2) Outdoor type transformer.

## 51. What are the types of transformers according to transformation?

Ans: (1) Step up transformer & (2) Step down transformer.

## 52. What do you mean by step up and step down transformer?

Ans: When the transformer transforms low voltage to high voltage it is called a step up transformer and when the transformer transforms high voltage to low voltage it is called a step down transformer .

## 53. what are the advantages of using a transformer in an a.c. supply?

Ans: The transformer has the following advantages :-

1. Voltage can be transformed from low voltage to high voltage for transmission and high voltage to low voltage for distribution without changing the frequency.
2. As the transformer has on rotating parts there is no frictional losses and were and tear of it .
3. A high voltage or an extra high voltage can be transformed easily by providing good insulation.
4. Its maintenance cost is low.
5. Practically it requires very little attention for its operation.

## 54. What is ideal transformer?

Ans: The transformer having an overall efficiency of 100 percent is called transformer.

## 55. What is the condition for an ideal transformer?

Ans : For an ideal transformer, Output =Input i.e. $V_2 I_2 = V_1 I_1$ or $E_2 I_2 = E_1 I_1$ or $I_2/I_1 = E_1/E_2 = N_1/N_2 = 1/k$ i.e. primary and secondary current are inversely proportional to their respective turns .

## 56. What do you mean by power transformer?

Ans: The transformer which are used up on transmission, lines for the transmission and distribution of relatively large quantities of energy are called power transformers.

## 57. What do you mean by transmission transformer?

Ans: The high capacity step up transformers , which are used in the generating station for onward transmission of electrical energy, are called transmission transformers.

## 58.What do you mean by distribution transformer?

Ans: When the transformers are used for distributing the energy from transmission lines and networks for local consumption and the secondaries are directly connected to the consumers load, they are called distribution transformers.

## 59. What do you mean by lighting transformer?

Ans: A transformer which is used to supply a distribution circuit having no motors connected to it , is called lighting transformer.

## 60. How does a transformer taken part for wide-spread popularity of a.c. system over d.c. ?

Ans: (1) High voltages of a.c. system can be obtained by using a transformer for transmission of electrical power.
(2) A.C voltages can be increased or decreased without any power loss by using a transformer.

## 61. Is there any difference of power if the secondary of the transformer is connected in star and delta respectively?

Ans: No. The power will be same .

## 62.Why is cooling necessary for power transformer ?

Ans: To prevent the damage to the coils and core due to excessive heat accumulated inside the transformers .

## 63. What are the sources of heart developed inside the transformer ?

Ans : Load current produces heat in the winding and eddy current produces heat in the iron core .

## 64. what are the important methods of cooling the transformer ?

Ans: The important methods of transformer cooling are as follows :-
1) Cooling by natural air.
2) Air forced type cooling.
3) Oil natural air natural type cooling.
4) Oil nature air forced type cooling.
5) Oil forced air forced type cooling.
6) Oil immersed water forced type cooling.
7) Oil forced water forced type cooling.

## 65. What do you mean by cooling by natural air ?

Ans: In natural air cooling system is effected by atmospheric air which passes inside the transformer through the holes provided on the casing of the transformer. Here convection process helps in cooling. This type of cooling is applied only in low KVA rating transformer.

## 66. What do you mine by air forced type cooling?

Ans: In air forced type of cooling, in addition to the natural air cooling air is forced inside the transformer by the arrangement of fans and circulated by introducing ventilating ducts.

## 67. What do you mean by oil natural air natural type cooling system?

Ans: It is most common system of cooling power transformer and distribution transformer. In this method transformer core and coils are immersed in a tank containing oil. The oil has both the property of cooling and insulating . The cooling tubes are fitted on the tank walls outside the tank through which the heated oil is circulated from the bottom to the top and cooled.

## 68.How is heat dissipated in oil cooled transformer ?

Ans: During running of transformer the oil in contact with core and coils becomes heated. The heated oil comes in contact with the casing and being lighter goes up through the cooling tubes where the heat is dissipated to the surrounding air. Finally cold having oil comes down and enters the tank again. This process continues in cyclic order so long the transformer is on running operation.

## 69. What is oil natural air forced type cooling?

Ans: In oil natural air forced type cooling method the oil cooling system is improved by the arrangements of fans which create air blasting.

## 70. What do you mean by oil forced air forced type cooling ?

Ans : In oil forced air forced type cooling system the oil is circulated through coolers which have cooling fans of exchange heat from oil to air.

## 71. What do you mean by oil immersed water forced type cooling ?

Ans: In oil immersed water forced cooling system the oil cooling method is further improved by the arrangement of tubes on the top of the transformer tank but below the oil level. Water is circulated through these tubes and cools the heated oil of the transformer.

## 72. What do you mean by oil forced water forced type cooling?

Ans: In the oil forced water forced type cooling water are circulated through the heat exchanger to exchange heat from oil to cooling water.

## 73. What are standard symbols used for describing different methods of cooling transformer ?

Ans: The following symbols are employed :-
A= Air cooling for dry type transformer.
N = Natural cooling by convection currents only .
G = Gas .
O = Oil immersed.
W = Water cooled .
F = Forced air or oil (oil not directed ).
S = Solid insulation used instead of mineral oil.
D = Forced (directed oil).
Practically cooling methods are expressed in two or more symbols together such as-
A.N – Air natural.
A.F – Air forced.
O.N – Oil natural.
O.F – Oil forced.
W.F – Water forced.
O.N.A.N – Oil natural air natural.
O.N.A.F- Oil natural air forced.
O.F.A.N – Oil forced air natural.
O.F.A.F –Oil forced air forced.
O.F.W.F – Oil forced water forced.
O.N.W.F – Oil natural water forced.
O.D.A.F – Directed oil forced and air forced.
O.D.W.F – Directed oil forced and water forced.

## 74. What do you mean by power oil? Given an example.

Ans: The oil used in electrical apparatus for cooling purpose as well as quenching of arc is called as power oil.

## 75. What is transformer oil?

Ans: Transformer oil is a variety of mineral oil obtained by refining crude petroleum which is an excellent insulator and its loss by evaporation is negligible small.

## 76. Why are power transformers filled with oil?

Ans: Power transformer are filled with oil to provide cooling by dissipating heat from coils and core to the outer casing and also to increase the dielectric strength of the insulation.

## 77.What do you mean by flash point and fire point of transformer oil?

Ans : The flash point of transformer oil is the ignition temperature of oil vapor i.e. the lowest temperature at which the oil gives off sufficient inflammable vapour to produce momentary flash on the application of a small flame . It is about 140$^{o}$C.

## 78 .What is the unit of measuring dielectric strength of the transformer oil?

Ans: The measuring unit of dielectric strength is Kilo-Volt /millimetre.

## 79 Which factors affect the dielectric strength of the transformer oil?

Ans : The moisture , dust and temperature affect the dielectric strength.

## 80. What is the effect of moisture on transformer oil?

Ans: The moisture decreases the dielectric strength.

## 81. What do you understand by the slugging in the transformer oil?

Ans: Sludging means the slow formation of semi-solid hydrocarbons in the transformer oil and deposition on the surface of the winding and the tank walls.

## 82.What do you mean by “Askarel” and where is it used?

Ans: “Askarel” means a group of synthetic liquids of high dielectric strength which overcomes the disadvantage of mineral insulated oil e.g. susceptibility to catching and propagate fire. This synthetic oil named as “Askarel” is used as an insulating and cooling medium in the electrical equipment where there is a necessity to reduce the risked of fire and explosion.

## 83. What are the fittings and accessories which should be provided on a transformer ?

Ans: The standard fittings and accessories - (1) Cable boxes, (2)Oil conservator, (3) Breather ,(4) Oil gauge, (5) Relief and Explosion vent,(6) Tapping switch, (7) Buchholz relay, (8)Radiator, (9)Bushings, (10) Oil level indicator, (11) Drain valve with plug, (12) Lifting hook, (13) Earthing terminal, (14) Nameplate, (15) Oil filling hole and cap, (16) Thermometer pocket & (17) Rollers.

## 84. What precautions do manufacturers take in the form of accessories to prevent transformer oil from deteriorating?

Ans: The following accessories are fitted as the precautionary measure against deterioration of transformer oil :- (1) Oil conservator, (2) Breather, (3)Buchholz relay & (4) Oil level indicator

## 85.What is a conservator ?

Ans: Conservator is a cylindrical tank placed on the top of the transformer connected to the main transformer oil tank by a short length pipe. It is partly filled with oil.

## 86. Why are oil conservators fitted to the transformer ?

Ans: Oil conservators are fitted to the transformers to take up the expansion and contraction of oil during running operation and also to observe the full oil filling of transformer tank through its oil level indicator.

## 87. Why is the conservation tank not completely filled with oil ?

Ans : During running condition the oil volume is increased with the increase of temperature . So the conservator tank is not completely filled with oil to absorb the increased volume of oil in its upper air space.

## 88. What is the lowest capacity of transformers on which the provision of conservators is obligatory ?

Ans: On transformers of capacities 50 KVA and above the provision of conservators is obligatory . On transformers of rating 40 KVA and below conservators shall not be fitted.

## 89.What is a breather ?

Ans: Breather is a small cylindrical chamber containing silica gel and connected to the top of the conservator by an air tight extended pipe. It allows moisture free air to the conservator tank.

## 90. What is the function of a breather ?

Ans: The function of a breather is to extract moisture from the ingoing air in order to prevent any contamination of the transformer oil by the moisture which is very much present in the open air.

## 91.What material is used to fill in the breather?

Ans: Moisture absorbing silicon gel or $CaCl_2$ is used to fill the breather which turns pink when it absorbs moisture . When fried again it gets its original form by removing its moisture contents.

## 92. What is emergency release ?

Ans : It is a large diameter explosion vent pipe fitted on the top of the tank of transformer projecting upwards and incorporating diaphragms on the mouth of it.

## 93. What is the purpose of explosion vent pipe on the transformer ?

Ans : If there is a large volume of gas formed within the tank due to short circuit fault in the winding the pressure is released from the transformer through the explosion vent pipe by rupturing the diaphragms mounted on the mouth of it.

## 94.How can the temperature of the hot oil be determined in a transformer ?

Ans : The temperature of the hot oil be determined with the help of a thermometer by putting its bulb inside a low pocket meant for thermometer situated near the top of the transformer tank where the oil is hottest.

## 95. What do you know about the temperature indicator of the transformer winding ?

Ans : The temperature indicator is nothing but a dial type thermometer fitted in such a way that its bulb is immersed in the region where the transformer oil is hottest and the bulb is associated with a small encircled heating coil connected to the is so designed that the bulb temperature is increased by an amount equal to the greatest

winding to oil temperature gradient and an appropriate value of temperature is indicated directly on the dial.

## 96. What are the causes of temperature rise in a transformer ?

Ans: Bad cooling and overloading are the causes of temperature rise in a transformer .

## 97. What are the losses in a transformer ?

Ans : The losses in a transformer are – (1) Core losses or iron or constant losses consisting of hysteresis and eddy current losses in the core and (2)Copper losses in the primary and secondary winding .

## 98. Is hysteresis objectionable ?

Ans: Since hysteresis is a kind of loss and affect the efficiency of the transformer, so it is always objectionable.

## 99.What are the factors on which the hysteresis loss depends?

Ans: The hysteresis loss depends on (1)the quality and constituents of the iron core, (2) the frequency and (3) the flux density.

## 100. Which iron loss is more in a transformer?

Ans: Hysteresis loss is more than eddy current loss in a transformer .

## 101. What is the main purpose of using silicon steel in the laminations of transformer core ?

Ans : To reduce the hysteresis loss.

## 102. Is there any variation of core flux between no load of full load ?

Ans: No, the core flux is always constant on every loads.

## 103. What are the functions of no load current in a transformer ?

Ans : No load current produces core flux and supplies iron losses and copper loss on no load.

## 104. What do you mean by the load current of a transformer ?

Ans : When a load is connected to the secondary side of a transformer the transformer is said to be on load and the current circulating through the secondary winding vai load is called the load current.

## 105. What do you mean by the primary current on load ?

Ans : When the load current ($I_2$) flows through the secondary winding a counterbalancing current ($I_{1'}$) is induced on the primary winding varying inversely to their respective turns . The total primary current on load ($I_1$) is equal to the vector sum of the primary counterbalancing current ($I_1$) and the no load current ($I_0$) which will be approximately equal to ($I_{1'}$) as $I_0$ is practically very small.

## 106. What will be the effect on primary said if the load current increases ?

Ans : Primary current increases with the increase in load current.

## 107. What is called the equivalent circuit of the transformer ?

Ans : The equivalent circuit of the transformer means the simplified corresponding transformer circuit modeling the behavior of a real transformer which represents the primary circuit as well as the secondary circuit in one circuit with suitable parameters for easy calculation with the addition of a shunt circuit to account for no load current.

## 108. What is called the approximate equivalent circuit of the transformer ?

Ans : The approximate equivalent circuit of the transformer means a slight modification of the equivalent circuit by transferring the shunt circuit to the primary terminal with close approximation affecting very little change in the result.

## 109. How do you calculate the equivalent resistance and reactance of the transformer referred to the primary and secondary ?

Ans: Equivalent resistance referred to primary $r_{1'} = r_1 + (N_1/N_2)^2 r_2$
Equivalent reactance referred to primary $x_{1'} = X_1 + (N_1/N_2)^2 x_2$
Equivalent resistance referred to secondary $r_{2'} = r_2 + (N_2/N_1)^2 r_1$
Equivalent reactance referred to secondary $X_{2'} = x_2 + (N_2/N_1)^2 x_1$
Where, $r_1$ and $r_2$ are the primary and secondary resistances respectively. And $x_1$ and

$x_2$ are the primary and secondary reactances respectively. $N_1$ and $N_2$ are the primary and secondary turns respectively.

110. A 50 KVA transformer has a turn ratio on ($N_1/N_2$ ) = 10. The impedance of the primary winding is 4+j7 ohms while that of the secondary winding is o.6 + j 1.1 ohms. What will be the impedance of the transformer when referred to the primary?

Ans : 64 + j 117 ohms.

111. What do you mean by resistance drop, reactance drop and impedance drop?

Ans : Resistance drop is the voltage drop in the resistance of the winding which is in phase with the current ($IR_1'$). Reactance drop is the voltage drop in the reactance of the winding which is in quadrature with the current($IX_1'$). Impedance drop is the vector sum of the resistance drop and the reactance drop.

112.What are the expression of find equivalent circuit ?

Ans: $V_1 = N_1/N_2\ [I_2(r_2'+jx_2')+V_2]$
$V_2 = N_2/N_1\ [V_1 - I_1\ (r_1' + jx_1')]$

113. Which voltage drop becomes larger in a transformer and why?

Ans : The reactance voltage drop becomes larger because of the large value of the inductive reactance of the transformer winding than its ohmic resistance.

114. How are core losses determined in a transformer ?

Ans : Core losses are determined by open circuit test. In this test normal voltage is applied on the low voltage side at rated frequency and since there is no load on the high voltage side the reading of the wattmeter connected on the low voltage side gives the no load loss or core losses of the transformer. High voltage side is kept open due to suitability of using metering arrangement in the low voltage side rather than high voltage side.

115. Why are copper losses neglected open circuit test?

Ans : Since no load current is very small and no load copper loss is very small in comparison with iron losses so the copper losses are neglected in open circuit test.

## 116. How are the copper losses determined in a transformer ?

Ans : Copper losses can be determined by short circuit test. In this test the low voltage secondary of the transformer is short circuited through an ammeter. At first a small voltage is applied to the high voltage primary side and secondary. Since the test voltage is very small in comparison to the normal voltage the core losses are negligible and the wattmeter connected on the primary side records the full load copper losses.

## 117. How is the copper loss affected by power factor ?

Ans : Copper loss varies inversely with the power factor.

## 118. In a short circuit test if the wattmeter reads 160 watts at 20 amperes of current what will be the reading of the wattmeter at 50 amperes of current ?

Ans : $W = 160 \times (50/20)^2 = 1000$ w

## 119. Why is a C.T used with the wattmeter when performing ashort circuit test on a transformer ?

Ans : Since the current coil of a wattmeter is capable of carrying about 5 amps therefore during short circuit test the current more than 5 amps will cause damage to the current coil. So for safety of the current coil of the wattmeter a current transformer is used.

## 120. What is impedance voltage ?

Ans: The impedance voltage of a transformer required to circulate rated current through a winding of the transformer when another winding is short circuited with the respective windings connected as for rated voltage operation. It is usually expressed in percent of the rated voltage of the winding in which the voltage is measured.

## 121. What do you understand by the efficiency of a transformer ?

Ans: The ordinary or commercial efficiency of a transformer is defined as the ratio of the output in watts to the input in watts. Since the efficiency is based on the power output in watts and not on volt-amperes, the power factor plays an important role in determining the efficiency.

Efficiency =(Output power)/(Input power) =($E_2$ $I_2$×P.F)/($E_2$ $I_2$×P.F+Losses)
= ($E_2$ $I_2$ cosØ)/($E_2$ $I_2$ cosØ+$W_1$+$I_1$' R1+$I_2$' $R_2$ )
Where $E_2$ =Secondary voltage.
$I_2$ = Secondary current.
$I_1$ =Primary current.
$R_1$ =Primary resistance.
$R_2$ = Secondary resistance.
$W_1$ =Iron losses.

## 122. What is the condition for the maximum efficiency of a transformer ?

Ans : The condition for the maximum efficiency of a transformer is (Iron losses =copper losses).

## 123. On what factors does the efficiency of a power transformer depend ?

Ans : The efficiency of a power transformer depends on the power factor of the load and the percentage of the load.

## 124. How does the efficiency of a power transformer depend on the load current ?

Ans : The efficiency increases with the increase in load current and reaches to maximum value and then decreases with further increase in load current .

## 125. A transformer has iron loss of 300 watts and full load copper loss of 500 watts. What will be the total loss of the transformer for maximum efficiency ?

Ans : 600 watts.

## 126. Under what condition of losses in a transformer maximum efficiency be obtained at 7/8 full load ?

Ans :When copper loss =64/49 × Iron losses.

## 127. Why is the efficiency of a transformer so high ?

Ans : Since the transformer is a static apparatus without any rotating parts there frictional loss. The only losses are iron losses end copper losses in the primary and secondary. Therefore, the efficiency of a transformer becomes so high about 95% to 98%.

## 128. What is the all day efficiency of a transformer ?

Ans : All day efficiency is defined as the ratio of energy output in 24 hours to energy input in 24 hours.

## 129. What is meant by regulation of a transformer ?

Ans : The regulation of a transformer is the difference between the no load and full load secondary voltages expressed in terms of the no load voltage with constant primary voltage.

## 130. How is the percentage regulation of a transformer is calculated ?

**Ans:** Percentage regulation $= \frac{No\ load\ voltage - full\ load\ voltage}{No\ load.voltage} \times 100\%$

$= \frac{E-V}{E} \times 100\% \quad = \frac{IR\cos\emptyset + IX\sin\emptyset}{E} \times 100\%$

When, R= Equivalent resistance in terms of secondary.
X =Equivalent resistance in terms of secondary.
E = Induced e.m.f. of secondary.
Φ = Phase difference between voltage and current.
I =Full load current.

## 131. What will be the effect of power factor and load on the regulation of a transformer ?

Ans : For lagging power factor of load current the regulation will be positive i.e. the secondary voltage will be reduced with the increase in the load current. For leading power factor of load current, the regulation will be negative i.e. the secondary voltage will be negative i.e. the secondary voltage will be increased with the increase in the load current.

## 132. Why is load test performed in addition to O.C. and S.C. test ?

Ans : Load test is performed not only to find the regulation and efficiency but also to measure the temperature rise of the transformer at different loads.

## 133. What is percentage impedance of a transformer ?

Ans : The impedance voltage is the vector sum of resistance voltage and reactance voltage. The percentage impedance is the value of impedance voltage expressed as a

percentage of rated voltage. Thus % Impedance = % Impedance voltage = $(V_1 Z_1')/V_1 \times 100$

Where, $I_1$ =Rated primary current.

$Z_1$ =Impedance with reference to the primary.

$V_1$ =Rated primary voltage.

## 134. What do you mean by tap changing ?

Ans : Tap changing is a process of changing the ratio of transformation by increasing or decreasing the number of active turns in one winding with respect to another winding for maintaining a constant voltage on a power system.

## 135. What are the types of tap changing arrangements ?

Ans : (1) OFF load tap changing & (2) ON load tap changing. Offload tap changing gears can be operated only when all the loads on the transformer are cut off. The voltage ratio of a transformer can be varied by about ± 10% by means of OFF load tap changing gears. ON load tap changing gears can be operated even when the load is on. The voltage ratio of a transformer can be varied by about ± 16% by means of ON load tap changing gears.

## 136. On which winding (H.V or L.V ) of a power transformer the tapings are normally provided and why ?

Ans : The tapings are generally provided on the high voltage side since the current loading is comparatively small on the high voltage side.

## 137. What is a diverter switch and why is it used ?

Ans: A diverter switch is a switching device having contacts with usually four separate make and brake units and arranged along with separate tap selectors. It is used in an on-load tap changer of a very large transformer to operate in rapid sequence and kept in a separate compartment to minimize oil pollution by carbon.

## 138. What is called bussing ?

Ans : Bussing is a type of porcelain or ebonite post insulator put on the top or side of the transformer tank through which connections are made to the external circuits.

## 139. What type of bushings is generally used according to the voltage of the transformer?

Ans : The types of bushing are as follows :- Procelain insulator bushing used up to 33KV. Oil filled bushings consisting of a hollow porcelain cylinder of special shape with a hollow tube-shaped conductor through its centre used for the voltages above 33 KV. Capacitor type bussing made to thick layers of bakelite paper alternating with thin garden layers of tin foil being covered by a porcelain rain shed and filled up with bitumen in the annular space between the rain shed and the bushing used in outdoor substation for the voltage above 33 KV.

## 140. What is an auto-transformer ?

Ans : An auto-transformer is a single winding transformer provided with a number of taps in which the single winding performs the functions of both the primary and the secondary.

## 141. What is the basic difference between distribution transformer and auto-transformer ?

Ans : The basic difference is that the distribution transformer has two separate windings and there is no electrical connection between the primary and the secondary while an auto-transformer has a single winding in which the primary and secondary are electrically connected.

## 142. What are the advantages of auto-transformers ?

Ans : (1) Being single winding transformer it requires less copper and becomes lighter in weight. (2) The cost is less than a two-winding transformer. (3) As it is provided with a large number of taps desired voltages can be obtained by adjusting suitable taps.

## 143. Why are the auto transformers not used for high voltages ?

Ans : If an auto-transformer is used in high voltage line a fault in the primary may affect the secondary side in which case the secondary load equipment may be damaged. So it is not used for high voltages.

## 144.What are the applications of auto-transformers ?

Ans : Auto transformers are used in case of low voltage and low transformation ratio e.g. starting equipment of three phase induction motor, control equipment of single phase and three phase electric locomotives, voltage control of power and lighting circuits, regulating transformers, boosters to raise the voltages in a.c. circuits etc.

## 145. What is a variable transformer ?

Ans : A variable transformer (named as Variac, Dimmerstat etc) is nothing but an auto-transformer having a continuously variable tap so that any output voltage desired could be obtained.

## 146. What are the different forms of connections used in primary and secondary windings in three phase transformer ?

Ans : The possible connections are as follows :- (1)Star/ Star, (2) Star/ Delta, (3) Delta/ Delta, (4) Delta/Star, (5) Open delta(Vee/Vee) & (6) Scott (Tee/Tee).

## 147. What is the application of Star/Star connection ?

Ans : Star/Star connection is best suited for small high voltage transformers with balanced load.

## 148. Where is Star/Delta connection applied ?

Ans : Star/Delta connection is applied to the transformer at sending end of the transmission line where the voltage is to be stepped up.

## 149. Where is Delta/Delta connection applied?

Ans : Delta/Delta connection may be used in large low voltage transformers where the insulation problem is not so urgent.

## 150. Where is Delta/Star connection applied ?

Ans : Delta/Star connection is generally applied in step-down distribution transformers to provide a 3-phase, 4-wire supply for three phase loads as well as single phase loads which may be balanced or unbalanced.

## 151. What are the advantages of having the primary winding connected in delta and the secondary winding in star ?

Ans : (1) An unbalanced four wire supply may be given. (2) The secondary neutral may be earthed or it may be utilized for given a four wire supply to provide both three phase and single phase loads.
(3) Third harmonic voltages are eliminated by the circulation of third harmonic current in the primary delta.

## 152. Where is open delta connection applied ?

Ans : Open delta connection is applied where the three phase load is too small and when one of the transformer in a delta-delta bank is disabled so that service is continued although at reduced capacity till the faulty transformer is repaired or a new one is substituted.

## 153. Where is Scott connection applied ?

Ans : Teaser or Scott connection is applied to supply three phase voltage from a two phase source and vice versa.

## 154. An 11KV/433KV, 500KVA, Δ/Y transformer is reconnected as Y/Y with high voltage side connected in star. What will be the rating of the new connection ?

Ans : 11.√3 KV/433V, 500 KVA.

## 155. What are the standard terminal markings on a transformer ?

Ans : High voltages terminals are marked with capital letters, N for neutral and A,B,C for three phases. Corresponding small letters are used for the low voltage terminals. In single phase transformers, the symbols are $A_2$ $A_1$ and $a_2$ $a_1$ in descending order from left to right.

## 156. What do you mean by phasing out of a 3-phase transformer ?

Ans : Phasing out of a 3-phase transformer means the detection of same phase terminals from primary and secondary winding.

## 157. How will you phase out a 3-phase transformer ?

Ans : For phasing out of transformers all phases are short circuited except a primary and a supposedly corresponding secondary. A small direct current is circulated in the primary and a voltmeter is connected across the secondary. A momentary deflection

of the voltmeter, when the primary current is made and broken, confirms that the two windings concerned belong to the same phase.

## 158. What are the symbolic vector groups for identification of 3-phase transformer ?

Ans : (1) Group – 1(Yy0, Dd0, Dz0, Zd0)
(2) Group – 2 (Yy6, Dd6, Dz6, Zd6)
(3) Group – 3 (Dy1, Yd1, Yz1, Zy1)
(4) Group – 4 (Dy11, Yd11, Yz11, Zy11).

## 159. What do you mean by the expression given in name plate of a transformer vector group Dy11 and impedance voltage 5% ?

Ans : A transformer with vector group symbol (Dy11) means a transformer having primary winding delta connected and secondary winding star connected and belonging to group-4 with +30$^o$ or-330$^o$ phase displacement between primary and secondary winding as referred to the standard counterclockwise vector rotation. Impedance voltage 5% means when 5% of normal voltage is applied across one winding it produces normal voltage is applied across one winding it produces normal full load current to flow through the secondary winding when its terminals are short-circuited.

## 160. what do you mean by parallel operation of transformers ?

Ans: To connect two or more transformers in parallel is called parallel operation of transformers.

## 161. What is the necessity of parallel operation ?

Ans : (1) To share the increasing demand of the load. (2) To continue the supply in case of any fault and periodical maintenance. (3) To get maximum operational efficiency.

## 163. What are the conditions to be fulfilled before paralleling two three phase transformers ?

Ans : The conditions are –

- The voltage transformation ratio must be same.
- The polarity must be same.

- The percentage impedance should be same.
- The phase sequence must be same.
- The vector diagram and the phase displacement must be same.

164. What will happen if the voltage ratios of the transformers running in parallel are not identical ?

Ans : The secondary voltage will be different for which a circulating current will flow through the two secondaries if the transformers are connected in parallel. This circulating current will act as the extra load without any useful work being done and cause heating of transformers even on no load.

165. Is it necessary for the ratings of two transformers to be the same for parallel operation ?

Ans : Not necessary but they should share the load in proportion to their ratings.

166. what will be the results of incorrect polarity and unequal percentage impedance of two transformers connected in parallel?

Ans : Incorrect polarity results in a dead short circuit and due to unequal percentage impedance the load sharing of two transformers will not be in proportion to their KVA ratings.

167. What are the effects of percentage impedance on load sharing ?

Ans : Since the percentage, impedance determines the voltage drop between no load to full load, with equal percentage impedances the transformers running in parallel will share the load in proportion to their capacities. The transformer with a lower percentage impedance will have a lesser voltage drop and therefore it will take a greater share of the load. On the contrary, the transformer with a higher percentage impedance will have a higher voltage drop and it will not take its due share of the load.

168. In parallel operation of two transformers, how will they share a load if their impedances are equal ?

Ans : Equally.

169. In parallel operation of two transformers if their ratings are equal but percentage impedances are different how will they share a load ?

Ans : If the ratings of the two transformers are equal but percentage impedances are different the loads will be shared inversely as the impedances of the transformers expressed in percent of normal voltage.

170. In parallel operation of two transformers if their ratings and also percentage impedances are different how do you calculate their load sharing ?

Ans: In the case of different percentage impedances for different ratings of transformers the percentage impedances have to be converted to the same basic KVA. Then the loads will be shared inversely as the converted impedances of the transformer.

171. In parallel operation of transformers of different capacities what should be the capacity of the smallest transformer with respect to the largest transformer in the group ?

Ans: The rated output of the smallest transformer in the group should not be less than 331/3 percent of the rated output of the largest transformer in the group.

172. How can you parallel a transformer in group-1 having vector group symbol Yy0, Dd0, Dz0 with one in group-2 having vector group symbol Yy6, Dd6, Dz6 ?

Ans: This can be done by altering the internal connection only of all the three secondary winding of any one transformer but not both.

173. How can you parallel a transformer in a group belonging to vector symbol Dy1 or Yd1 or Yz1 with one in another group belonging to voltage symbol Dy11 or Yd11 orYz11 ?

Ans: This can be done by transposing two of the external high voltage connections of one of the transformers and also the two corresponding low voltage external connections. The change of any of the internal connections is not necessary.

174. Is it possible to operate a transformer of groups 1&2 in parallel with a transformer of groups 3&4 listed below ? If not ,why ?

| Group-1 | Group-2 | Group-3 | Group-4 |
|---|---|---|---|
| Yy0 | Yy6 | Dy1 | Dy11 |
| Dd0 | Dd6 | Yd1 | Yd11 |
| Dz0 | Dz6 | Yz1 | Yz11 |

Ans : Transformer in groups 1 & 2 can not be operated in parallel with those in groups 3 & 4 because there is an inherent phase angle difference which can not be corrected.

## 175. What will be the effect of the change in voltage on the performance of a transformer ?

Ans : Iron loss varies approximately as square of the voltage. Copper loss and regulation vary inversely as square of the voltage. Iron temperature increases and copper winding temperature decreases with increase in voltage & vice versa. Efficiency at fractional load decreases and that at full load increases with increase in voltage and vice versa.

## 176. What will be the effect of a change in voltage on the performance of a transformer ?

Ans: Iron loss and temperature will vary inversely as the frequency. Copper loss is independent of frequency. So total loss increases with the decrease in frequency. Accordingly, efficiency decreases with the decrease in frequency. The regulator at low power factor decreases with a decrease in frequency and vice versa.

## 177. Why does a transformer require protection ?

Ans : A transformer requires protection for the following hazards :-
Overloading and short circuit fault. Internal fault – (a) Loose contact, (b) Short circuit between adjacent turns & (c) Earth fault in the winding or leads. External hazards – (a) Lightning surges & (b) System overvoltage.

## 178. What are the protective equipment required for a large transformer ?

Ans : The protective equipments are i) Buchholz relay,
ii) Over current relay,
iii) Differential relay,
iv) Earth leakage relay &
v) Lightning arrestor.

## 179. What is the reason for gas formation in a transformer ?

Ans : When the heat produced by any internal fault inside the tank is intensified then it decomposes the oil and liberates gas.

## 180. What is the function of Buchholz relay ?

Ans : The function of Buchholz relay is to actuate an alarm in case of small amount of gas or trip the circuit breaker in case of excessive gas by making use of that gas liberated inside the transformer due to any internal fault. It also detects the accumulation of air on account of low oil level or faulty oil pumps.

## 181. What is the lowest capacity of the transformer on which the provisions of gas pressure type protection is obligatory ?

Ans : On transformers of rating 1000 KVA and above the provision of gas pressure type protection with Buchholz relay is obligatory as per I.E. rules.

## 182. Where is Buchholtz relay installed ?

Ans : The Buchholz relay is installed in the pipe connecting the transformer tank and the oil conservator.

## 183. what should be the angle of inclination of the connecting pipe of the Buchholz relay with the horizontal plane ?

Ans : The angle of inclination of the axis of the pipe with horizontal plane should be $10^o$ to $11^o$ .

## 184. What should be the size of the connecting pipe of Buchholz relay ?

Ans : The nominal pipe bore diameter should be 25 mm for transformers up to 1000 KVA, 50 mm for transformers exceeding 1000 KVA but up to 10000 KVA and 80 mm for transformers exceeding 10000 KVA capacity. The length of the pipe of the lower section between relay and transformer tank should be more than 5D and that of the upper section between relay and conservator tank should be more than 3D (where D is the internal diameter of the connecting pipe).

## 185. What is the function of overcurrent relay ?

Ans : The function of overcurrent relay is to protect the transformers against overload and short circuit.

## 186. What is the lowest capacity of the transformers on which the provision of differential protection is obligatory ?

Ans : On transformer of capacity 10 MVA and above the provision of differential protection against incipient fault is obligatory as per I.E rules.

## 187. What do you mean by instrument transformers and why are they used ?

Ans : An instrument transformer is a transformer designed to represent the condition of current or voltage and of phase position in the primary circuit with acceptable accuracy in the secondary circuit. It changes the voltage or current in a power circuit to values which render them convenient for measurement. Instrument transformers are used in circuits of high voltage or large current to operate measuring and recording instruments and protective relays.

## 188. How do the instrument transformers differ from the power transformer ?

Ans : The instrument transformers differ from the power transformers mainly by their volt-ampere rating. The instrument transformers have a very low rating of only a few tens of VA e.g 10VA, 50VA, 100VA etc. whereas the power transformers have high KVA rating e.g 25KVA, 200KVA, 500KVA, 1000KVA even up to 1000MVA.

## 189. What are the types of instrument transformers ?

Ans : There are two basic types, namely, current transformers and potential transformers.

## 190. What is called true ratio of an instrument transformer ?

Ans : The true ratio of a current or a potential transformer is the ratio of r.m.s primary current or voltage, as the case may be, to the r.m.s secondary current or voltage under specified conditions.

## 191. what is called marked ratio of an instrument transformer ?

Ans : The making ratio of a current or a potential transformer is the ratio of the primary current or voltage, as the case may be, to the secondary current or voltage as given on the rating plate.

## 192. What is called ratio correction factor ?

Ans : The ratio correction factor is that factor by which the marked ratio of a current or a potential transformer must be multiplied to obtain the true ratio. So R.C.F = (True ratio)/(Marked ratio.)

## 193. What are the types of current transformers ?

Ans : The two types are (1) Air-cooled current transformers and (2) Oil immersed current transformers.

## 194. What are the types of current transformers according to the magnetic circuit ?

Ans : There are three types of current transformers in common use according to the magnetic circuit, namely, “Ring type”, “Core type” and “Shell type”.

## 195. How are the terminals marked in a current transformer ?

Ans : The primary terminals are marked by the letters ‘M’ and ‘L’ for main & load ends respectively and the secondary terminals by the same letters enclosed in a circle i.e. (M) and (L).

## 196. What are the errors in C.Ts. ?

Ans : There are two of errors, e.g. ratio error and phase difference error. The ratio error is the amount by which the secondary current differs from exact proportionality to the primary current and is expressed as a percentage of the rated secondary current. The phase difference error is the angle by which the secondary current differs in phase form the primary current and is expressed in minutes of arc.

## 197. At what load the phase angle error of a c.t. will be negative ?

Ans : At the load of very low power factor the phase angle error will be negative.

## 198. why are the errors of a C.T. generally greater with relatively small loads than at rated full load?

Ans: The errors are generally increased with relatively small loads than at rated full load due to greater proportion of exciting current to the reversed secondary current .

## 199. If the core of a C.T. is permanently magnetized how will you demagnetize it ?

Ans: To demagnetize the core of a C.T. a variable resistance of high value is connected across the secondary terminals and with full rated current flowing through the primary the resistance is gradually reduced to zero.

## 200. What is meant by "turn compensation" and why is it done ?

Ans : Turn compensation means to keep the number of secondary turns one or two less than that obtained by normal ratio and it is done to reduce the current ratio error.

## 201. How are the ratio error and the phase error of a C.T. improved ?

Ans : The errors of a C.T. can be improved by the following methods – By using high permeability and low loss magnetic material of the core. By reducing the length of the flux path in the core and increasing the area of the path with all joints reduced to a minimum or avoided altogether. By increasing the primary ampere-turns. By using one or two turns less in the secondary winding. By reducing the internal secondary burden as far as possible. By keeping the connected burden on the secondary as small as possible. By specifying the rated burden as near to the actual burden as possible.

## 202. What are the categories of current transformers and their applications ?

Ans : There are two categories of C.Ts, e.g, Measuring current transformers are used with ammeters, wattmeters, KVA meters, KWH meters, power factor meters etc. Protective current transformers are used with over current relays, earth fault relays, differential protection, impedance protections etc.

## 203. Which instruments transformer is used in a Tong tester ?

Ans : Current transformer.

## 204.why is only one turn of very thick conductor used in the primary of a current transformer ?

Ans : In the primary of a current transformer only one turn of very thick conductor having negligible resistance is used not only for reducing the induced voltage but also for reducing the voltage drop to nearly zero. As large current flows through the primary and small current as per current ratio flows through the secondary if one turn is selected for primary the secondary turns will be smaller.

## 205. What materials are used for the core of a C.T. & why ?

Ans : In order to reduce the ampere turns required the core of a C.T. must have high permeability, small iron loss, a low flux density and low phase difference error. Two suitable materials are Silicon steel having the maximum permeability of 4500 at flux densities in the neighborhood of 5000 lines/cm$^2$ with phase difference error of 1.5 to 2$^o$ . Mumetal, a nickel-iron alloy containing a small amount of copper having the maximum permeability of about 80000 at flux densities of about 3500 lines/cm$^2$ with phase difference error of below 0.5$^o$. As the nickel-iron alloy reduces the magnetizing current and core loss to about 10% of that of the corresponding silicon steel values, Mumetal is best suited for the core of a C.T. where a high degree of accuracy is desired.

## 206. What is the effect of saturation on the performance of C.T. ?

Ans : On reaching saturation the transformer ratio is affected resulting in an increase of exciting current due to reduction in magnetizing inductance and introducing a phase error in the C.T. deviating primary and secondary current from their desired phase opposition i.e. 180$^o$. The increase in exciting current causes the reduced secondary output and hence reduced the speed of over current relays. The saturation of C.T. disturbs the balance in differential relays thus affecting the stability of protection. Therefore the effect of saturation makes the operation of the C.T. unsatisfactory.

## 207. What are the basic differences between current transformers and potential transformers ?

Ans : (1) A C.T . is connected in series with the line whereas a P.T. is connected across the supply lines. (2) In a C.T the number of turns is inversely proportional to the current but in a P.T. the number of turns is directly proportional to the voltage. (3) In a C.T. the primary has one or more turns of heavy conductor and the secondary has a large number of turns of thin wires whereas in a P.T. the primary has a large number of turns of thin wires and the secondary has a few number of turns of comparatively thick wires.

## 208. What are the types of potential transformers ? State the applications of potential transformers ?

Ans : There are two types of potential transformers. E.g. Electromagnetic type potential transformer, in which primary and secondary windings are used on the magnetic core like usual transformers. Capacitor type potential transformer, in which a series capacitor group is used with an auxiliary voltage transformer. The primary voltage is applied to the series capacitor group and the voltage across one of the capacitor is taken to the primary of the auxiliary voltage transformer. Then the secondary of the auxiliary voltage transformer is finally taken to the metering and/ or control panel. Potential transformers are used with voltmeter, KWh meter, power factor meter, frequency meter, protective relays for the purpose of measurement and protection. Electromagnetic type potential transformers are used upto 66KV lines while capacitor type potential transformers are used for the voltages 66KV and above.

## 209. How are the terminals of a potential transformer marked ?

Ans : In single phase potential transformer the primary terminals are marked with letters $V_1$,$V_2$and the secondary terminals bearing similar markings are encircled. In the case of tapped windings the primary terminals are marked with the letters $V_1$, $V_2$,$V_3$,$V_4$ and the secondary terminals bearing similar markings are encircled. In three phase star/star potential transformer the primary terminals are marked with the letters A B C N and the secondary terminals are marked with similar letters but encircled. The terminal marked with the letter 'N' is the neutral.

## 210. What do you mean by the "ratio error" and "phase difference error" of potential transformer ?

Ans : The amount by which the secondary voltage of a potential transformer differs in magnitude from exact proportionality to the primary terminal voltage is called the ratio error of a potential transformer. The angle by which the secondary voltage of a potential transformer differs in phase from the primary terminal voltage is called the phase difference error of a potential transformer.

## 211. What will be the effect of open circuiting the secondary of a C.T. ?

Ans : If the secondary of a C.T. is open circuited the whole current in the primary becomes magnetizing current & a dangerously high voltage will appear across the secondary which may cause insulation breakdown, damage to the magnetic property

of its iron core, overheating and also death to life. For these reasons open circuiting of a C.T. should never be permitted.

212. What is the usual limit of current and voltage above which C.T. and P.T. are generally used ?

Ans : Current limit -50 amps & voltage limit-750 volts.

213. what are the standard ratings of instrument transformers ?

Ans : The standard secondary rating of P.T. is 110 volts and that of C.T. is 5 amps.

214. What do you mean by 'Burden' of a P.T. or C.T. ?

Ans : The maximum load in volt amperes (VA) which may be applied across the secondary terminals of P.T. or C.T. is known as 'burden' of a P.T. or C.T. It depends upon the number of instruments or relays connected and their individual ratings.

215. What should be done before disconnecting an ammeter from the secondary of a C.T. ?

Ans : The secondary winding of a current transformer should always be short circuited before disconnecting an ammeter.

216. What is the basic difference between a power transformer and a potential transformer according to the operating condition ?

Ans : Though the primary winding in both power transformer and potential transformer are energized continuously at a substantially constant voltage, the secondary of a power transformer is connected to a load which may vary between zero and the maximum which the transformer is capable of carrying whereas the load or burden connected to the secondary of a potential transformer does not vary as a rule. The permissible voltage drop in potential transformer between zero and maximum burden is much smaller than the corresponding limit in a power transformer.

217. What precautions do you observe when working with instrument transformers ?

Ans : When working with C.T. the secondary should be short circuited and grounded. No fuse should be inserted in the secondary circuit of C.T. because fuse may blow

and produce an open circuit. When working with P.T. the secondary should never be short circuited. One end of the secondary should be grounded and fuse should be provided both on the primary and the secondary circuit. one end of the secondary should be grounded and fuse should be provided both on the primary and the secondary circuit.

218. How are the earthing arrangements made for the following :-
a) 132 KV or 220 KV generator transformers,
b) 11Kv or 33KV substation transformers,
c) 11KV/0.433KV distribution transformers,
d) Transformer with delta winding &
e) Instrument transformer ?

Ans : (a) The metallic frame is earthed by two separate and distinct connections with earth. The neutral point is directly earthed by not less than two separate and distinct connection with earth each having its own electrode thus permitting the grading of insulation in the transformer form the terminal end to the neutral point.
(b) The frame is earthed by two separate earth electrodes as above but for neutral earthing resistance earthing not less than two numbers may be used to limit the fault current which is expected to be too high in this case.
(c) The frame is earthed by separate earth electrodes and the neutral of the secondary star winding is erathed by not less than two nos separate and distinct connections directly to earth having its own electrode.
(d) The frame is earthed by two separate earth electrodes . An earthing transformer may be used to get the star neutral point which may be earthed by two separate connections to the earth either directly or through a resistance if desired.
(e) Cases or frames of instrument transformers, the secondary winding of current transformer and one point of the secondary winding of potential transformer shall be earthed which may be done by connection to the earth bus.

219. What is earthing transformer & where is it used ?

Ans : An earthing transformer is a transformer which is intended primarily for the purpose of providing a neutral point for grounding purposes. It may be a two winding unit with a delta connected primary and a star connected secondary or a single winding three phase auto-transformer with windings in interconnected star or “Zig-Zag”. When the necessity arises for earthing the neutral of one part or another of on interconnected transmission or distribution network at a place where no natural

neutral point is available, an earthing transformer is used for that purpose. The neutral point of the earthing transformer is connected to earth directly or through a current limiting impedance which the terminals are connected to the three phase lines.

## 220. What is tertiary winding ? what are its function ?

Ans : Tertiary winding is a delta connected additional auxiliary winding used with three phase star-star connected transformers which is wound on the same core and separated from primary and secondary main winding . The function of the tertiary winding are as follows :-
(i) It prevents the oscillation of the neutral point i.e stabilizes the neutrals of both primary and secondary winding by reducing third harmonic voltage components.
(ii) It permits the transformation of unbalance three phase loads or single phase loads only by limited the voltage unbalance.
(iii) It can supply an auxiliary load at low voltage mainly for station loads.

## 221.What are the important aspects related with the installation &operation of power transformers ?

Ans : The important aspects are as follows :-
i) Interchangeability of transformers & spares from the maintenance point of view.
ii) Installation layout mentioning safe clearance from the neighbouring equipments & structures.
iii) Necessary arrangements for drainage of oil.
iv) Provision for isolation the transformer for maintenance work.
v) Safety precautions.
vi) Protections of transformer.
vii) Grounding of neutral point, tank, cores etc.
viii) Provision for surge protection.
ix) Provision fire fighting arrangements.

## 222. What important things need to be ensured in installing a 11000/433 volts, 500KVA distributing transformer indoors ?

Ans :The most important thing to be ensured with transformer installed indoors is proper ventilation. The level of the transformer base should be higher than the highest flood and storm water level of that area. The transformer should be kept well away from the wall. The site should be so chosen that it is free from chemical fumes

&damp. For indoor installation, the air inlets and outlets shall be of adequate size and so placed as to ensure proper air circulation for the efficient cooling of transformers. Minimum clearance between the highest point of the conservation tank and ceiling of the transformer room should be sufficient to remove the mounting on the transformer link the conservation. The transformer should be so installed that severe vibrations are not transmitted to its body.

## 223. What are the tools and equipments required for the erection of a transformer ?

Ans : General list of tools and equipments :-
i) Lifting equipment (crane or chain pulley block along with wire rope slings, D-shackles etc).
ii) Vacuum oil purifier.
iii) Vacuum pump.
iv) Oil storage tank.
v) Pressure vacuum gauge.
vi) Oil testing apparatus.
vii) 2500/1000 volts meggar.
viii) Voltmeter, milli-ammeter, low p.f. wattmeter (voltmeter of 0-500V, 0-100V, 0-5V range).
ix) AVO meter.
x) Set of spanners.
xi) Set of drum opener, crowbar, pipes, hammer etc.
xii) Set of screw drivers, cutting pliers, screw spanners and pipe wrench.
xiii) Clean cotton cloth and cotton waste.
xiv) Electric hand lamp.
xv) 12 mm venyl hose of approximate 10 meters length for using as an oil level indicator during erection.
xvi) Painting brush.
xvii) P.V.C. wires for all necessary electrical connections during testing.

## 224. What should be the arrangements for proper ventilation of transformer installed indoor in an enclosed room ?

Ans : For proper ventilation the transformer should be kept well away from the to make free movement of air round all the four sides. To ensure proper air circulation for the efficient cooling of the transformer a minimum area of 1 sq. meter for inlet per 1000 KVA should be provided as near the floor as possible and a minimum area

of 2 sq. meters for outlet per 1000 KVA should be provided in the opposite side of air inlet as high as the building allows to enable the heated air to escape readily and be replaced by cool air.

225. In a transformer cubicle proper air circulation can not be achieved due to wall blocking. What kind of arrangement is to be provided in the cubicle for efficient dissipation of heat form the transformer ?

Ans : In addition to the provision of proper air inlet and outlet fans are provided extra for cooling the transformer.

226. What is minimum recommended spacing between the walls and transformer periphery in case (a) wall on one side, (b) wall on two sides, (c) wall on three sides and (d) wall on all sides as in an enclosed room ?

Ans : (a) 0.5 meter (b) 0.75 meter (c) 1.00 meter & (d) 1.25 meters.

227. What should be the minimum clearance between the highest point of the conservation tank and the ceiling of the transformer room ?

Ans : Not less than 0.5 meters.

228. What arrangement should be provided where two or more transformers are installed side by side ?

Ans : If two or more transformers are installed side by side they shall be separated by fire separation walls. Fire separation walls are deemed to be adequate from fire safety point of view.

229. What type of oil draining arrangement should be provided for a transformer having more than 2000 liters of oil ?

Ans : Oil soak pit filled with sand and gravels should be provided where the oil capacity exceeds 2000 liters of oil. The capacity of the oil soak pit shall be such that to soak the entire oil content of the transformer.

230. When is it necessary to dry out a transformer ?

Ans : As a result of tests carried out if the presence of moisture is indicated or the oil does not withstand the tests as recommended by the manufacturer or the insulation resistance readings are not satisfactory it shall be necessary to dry out a transformer.

## 231. What are the methods of drying out transformer core and coils ?

Ans : i) Drying with core and coils in oil by short circuit method.
ii) Dering with oil removed by external heat.
iii) Drying under vacuum.

## 232. What are the precautions to be taken when drying out of transformers ?

Ans : (i) The transformer should never be left unattended during drying out. This must be done under continuous and competent supervision.
(ii) The temperature should be carefully observed which should not exceed $90^{o}C$.
(iii) Only spirit type thermometer should be used.
(iv) No naked lights and flames should be allowed in the vicinity of a transformer being dried and the fire fighting equipment should be kept ready during the dry-out period.
(v) Low voltage hand lamp should be used for inspection of the tank from inside.

## 233. How will you ascertain the completion of drying out ?

Ans : During the drying period the measurement of insulation resistance should be taken every 2 hours at practically constant temperature. In the course of time when the insulation resistance remains constant for a period of 24 hours at a value greater than its suggested value the drying out process is to be taken as completed.

## 234. What do you mean by reactivation of transformer oil ?

Ans : Reactivation of transformer oil means the reconditioning of deteriorated oil to make it active again by purifying and drying transformer oils.

## 235. What are the different methods of purifying and drying out transformer oils ?

Ans : The different methods are
i. Filter press.
ii. Centrifugal oil purifier.
iii. Streamline filter.

## 236. What do you mean by neutralization value of transformer oil ?

Ans : The neutralization value also known as total acidity is the measure of free organic and inorganic acids present together in the oil which is expressed in terms of

the number of milligrams of potassium hydroxide required to neutralize the total free acids in one gram of the oil.

## 237. What is the maximum limit of acidity of serviceable transformer oil ?

Ans : 0.40 mg of KOH per gram of oil.

## 238. what is the effect of the presence of moisture and impurities in oil ?

Ans : The presence of moisture and impurities in the oil decreases the dielectric strength of the oil and also reduces the insulation resistance of the transformer.

## 239. What is the effect of presence of air bubbles in the transformer oil ?

Ans : When air remains in the oil the oxygen reacts on the cellulose of the insulation and the decomposition products of the cellulose lead to the formation of organic acids soluble in oil and also sludge which blocks the free circulation of the oil.

## 240. What is the effect of varnishes on the transformer oil ?

Ans : varnishes of the oxidizing type readily reacts with the transformer oil and precipitate sludge on the windings. But synthetic varnishes having acid inhibiting properties generally delay the natural formation of acid and sludge in the oil.

## 241. What is the effect of slackness of windings in the transformer ?

Ans : Slackness of winding may cause a failure due to repeated movement of coils which may be the conductor insulation at some places and lead to an inter-turn failure. The coils may also get displaced under load conditions or momentary short circuit which may cause electric and magnetic unbalance and produce even greater displacement.

## 242.How do you detect the presence of water in transformer oil ?

Ans : It can be detected by crackle test. For this test some quantity of oil is heated in a test tube over a silent flame or a red-hot metal rod is dipped in a cup containing oil. Any cracking sound, in either case, indicates the presence of water.

## 243. What type of gasket is generally used in transformer and why ?

Ans : Neoprene bonded cork gasket is generally used in between two metal surfaces in transformer assembly which is squeezed by tightening nut bolts of the flanges to make a leak-proof air-tight and oil tight joint.

## 244. What are the possible causes for abnormal noise in a transformer ?

Ans : External noise is caused by a loose nut-dolt on the transformer cover and its accessories and internal noise is caused by slackness of winding or capacitance discharge.

## 245. What are the causes of loosening of coils in the transformer ?

Ans : The causes are as following :
(i) Alternate heating and cooling.
(ii) Uneven electromagnetic forces due to defective construction.
(iii) Electric and magnetic unbalance due to abnormal load conditions or momentary short circuit.

## 246. What will happen if a transformer is connected to a d.c. source ?

Ans : If a d.c. supply is connected across the primary winding it will produce a steady flux in core and therefore it will not induce any voltage. Secondly, the transformer winding has more reactance and very little resistance. Since the reactance has got no effect with d.c. voltage the transformer will take high current due to low ohmic resistance and absence of self induced counter e.m.f. As a result the winding will be burnt out without transforming the electrical energy.